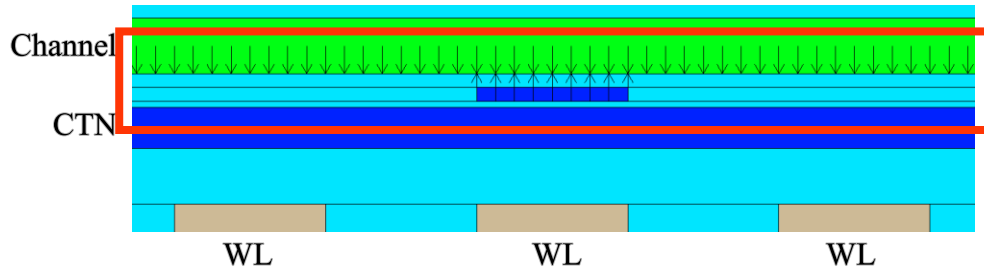


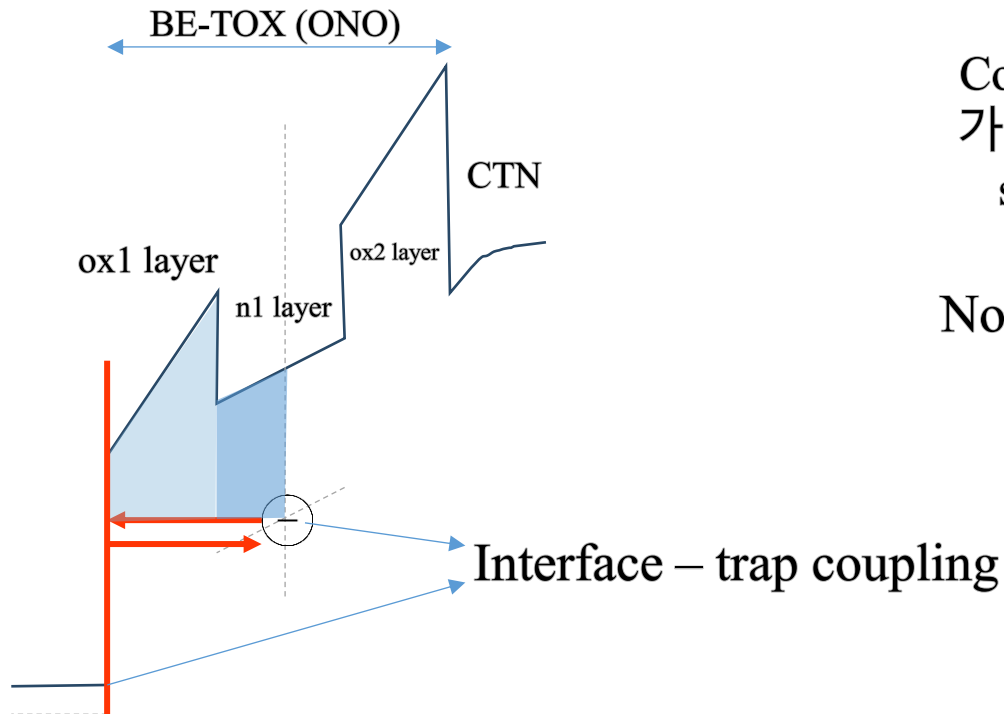
Modeling of Detrapping electron in BE-TOX by Elastic Tunneling Emission

20241129 진명

NonLocal Tunneling in TCAD



Nonlocal mesh
: interface – mesh(trap) coupling



Coupled trap – interface 관계
가 SRH theory와 같은 steady
state 상태에서 이루어짐

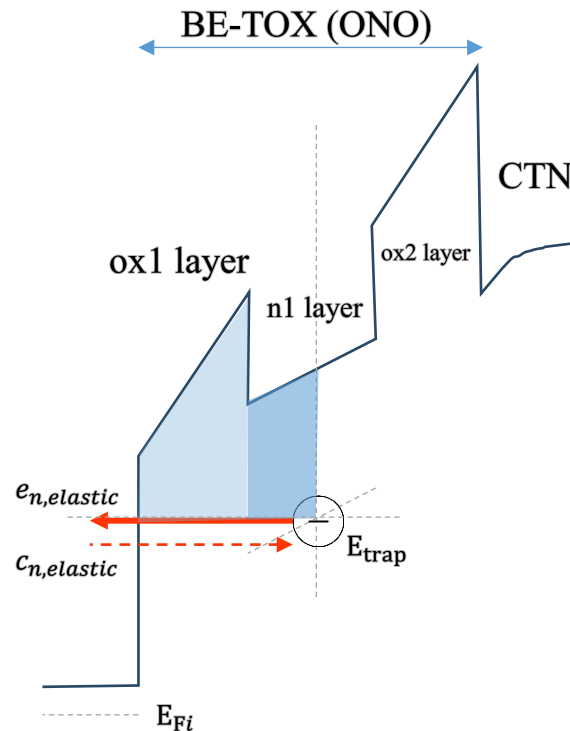
+

Nonlocal tunneling = Elastic
+ phonon-assisted

Elastic Tunneling Capture & Emission

$$c_{n,\text{elastic}} = \frac{2\pi}{\hbar} |E_T|^2 V_T \cdot \frac{\sqrt{2}}{\pi^2 \hbar^3} \sqrt{m_{\perp}^*} m_{0\parallel} \sqrt{E_{\text{trap}} - E_{c0}} \frac{1}{1 + \exp\left(\frac{E_{\text{trap}} - E_{Fi}}{kT}\right)} |TC|$$

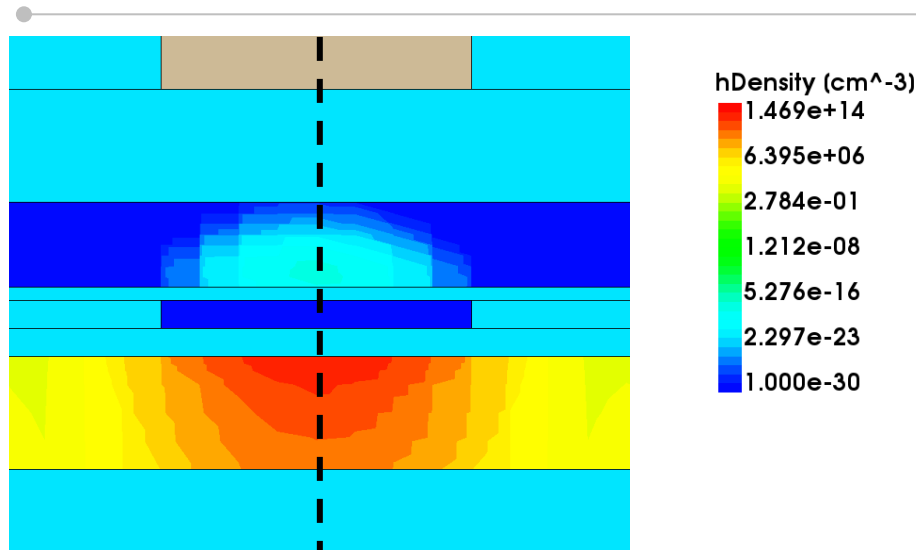
$$e_{n,\text{elastic}} = \frac{2\pi}{\hbar} |E_T|^2 V_T \cdot \frac{\sqrt{2}}{\pi^2 \hbar^3} \sqrt{m_{\perp}^*} m_{0\parallel} \sqrt{E_{\text{trap}} - E_{c0}} \frac{1}{\exp\left(-\frac{E_{\text{trap}} - E_{Fi}}{kT}\right) + 1} |TC|$$



Since $E_{\text{trap}} \gg E_{Fi}$, emission $\gg$ capture

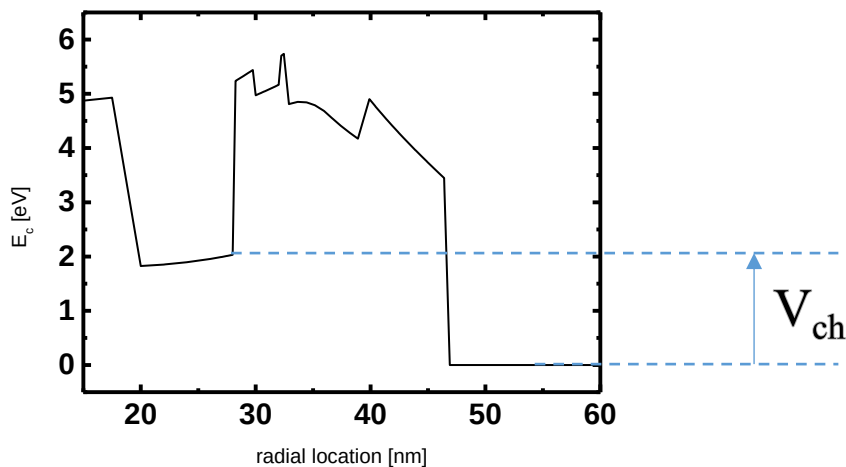
Trap – to – Band emission occurs

Channel Potential



CTN programmed electron
forms negative voltage

>> hole accumulation at
channel



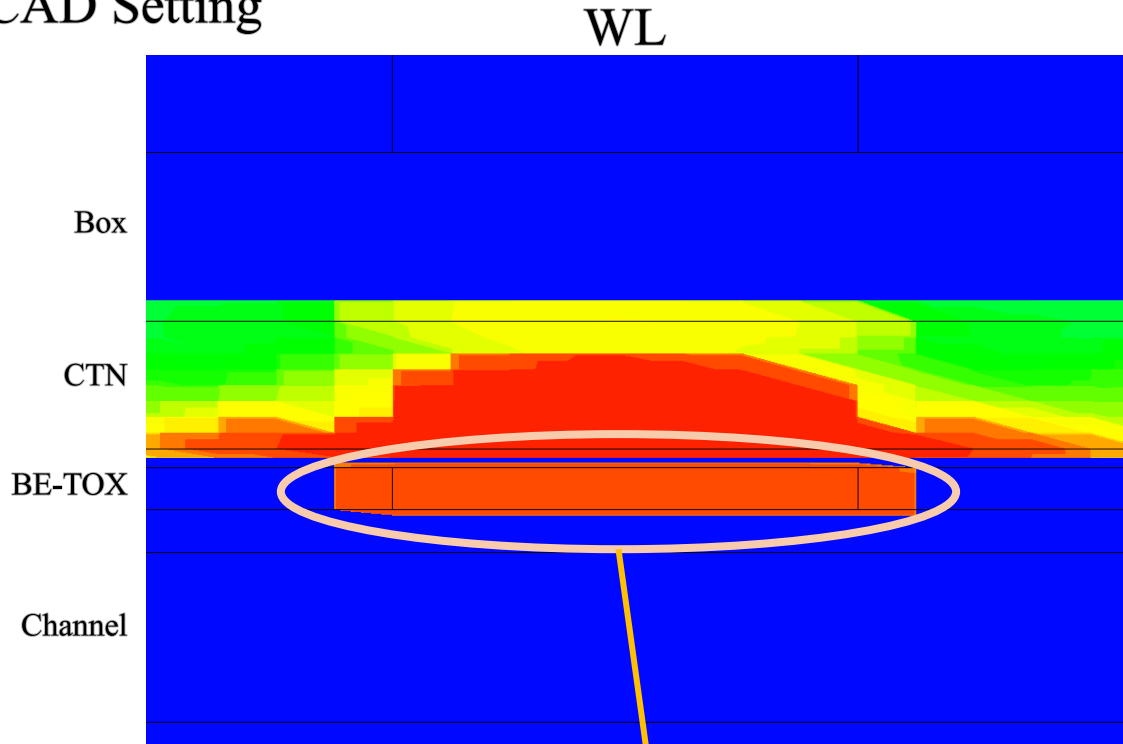
Negative channel potential bias!

20V single pulse 1us program:
 $V_{ch} = -1.9V$

(Can be specified for certain PV level)

TCAD & Modeling Method

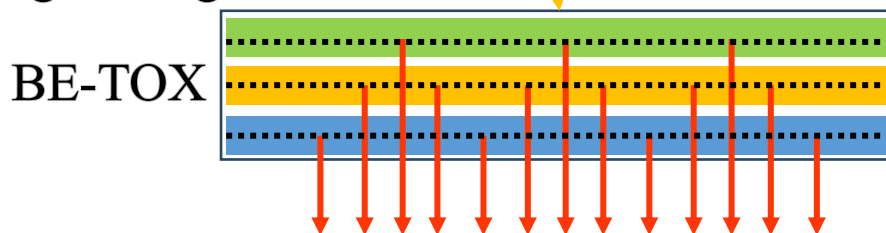
TCAD Setting



CTN CCS Nullification (~ 0)
> Ideal electron retention

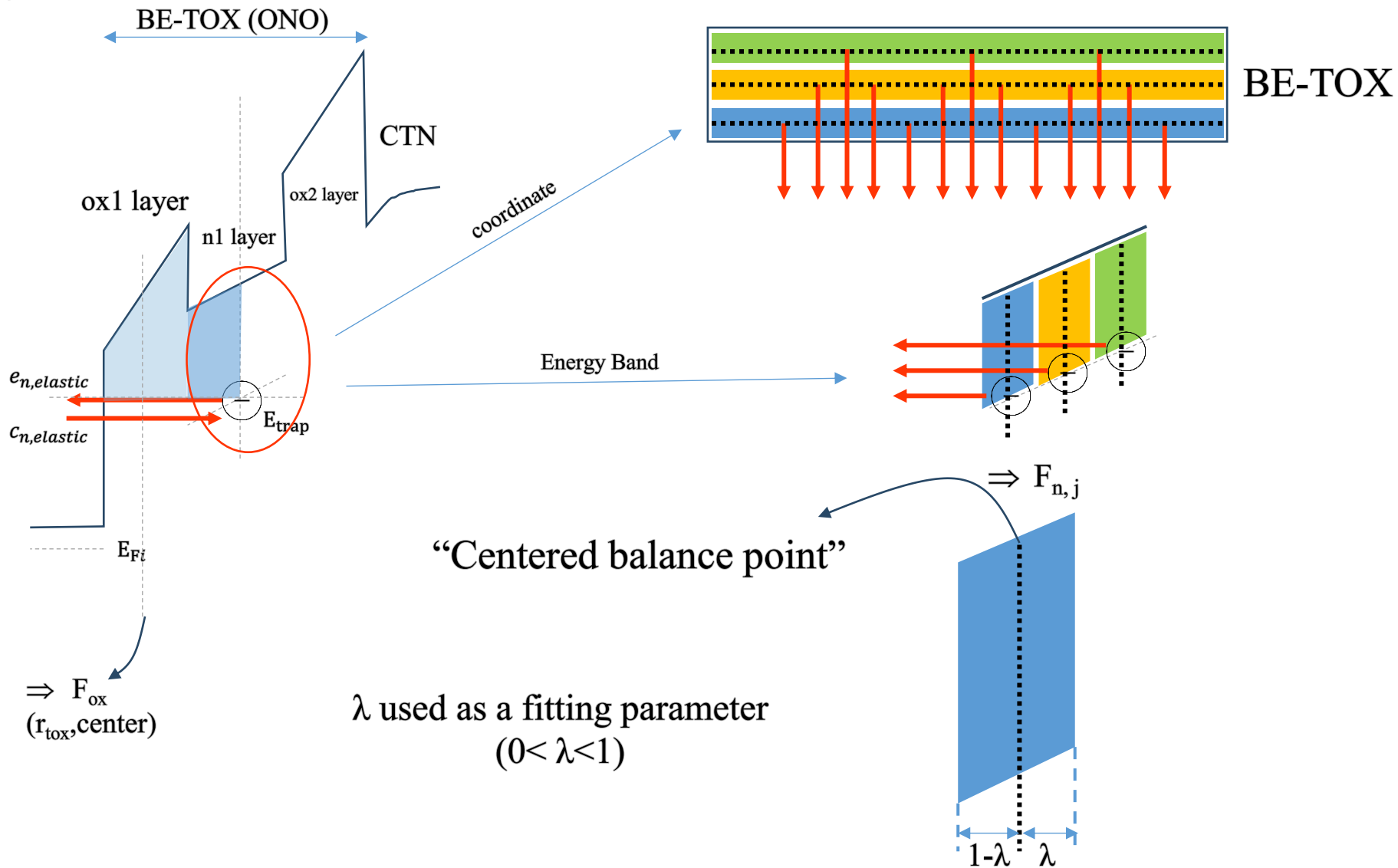
BETOX oxide envelope
> No lateral spreading occurs

Modeling Setting



Modeling designed
with vertical segments

Modeling Method of Coordinate Segment



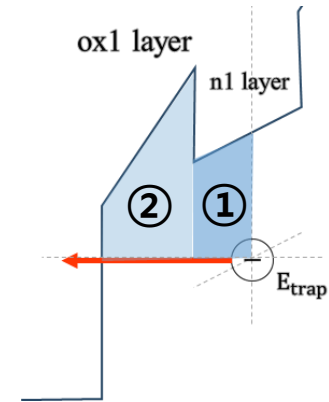
Transmission Coefficient

$$C_{n,\text{elastic}} = \sum_j \frac{2\pi}{\hbar} |E_T|^2 V_T \cdot \frac{\sqrt{2}}{\pi^2 \hbar^3} \sqrt{m_{\perp}^*} m_{0\parallel} \sqrt{E_{\text{trap}} - E_{c0}} \frac{1}{1 + \exp\left(\frac{E_{\text{trap}} - E_{Fi}}{kT}\right)} \quad \text{[TC}_j\text{]}$$

$$e_{n,\text{elastic}} = \sum_j \frac{2\pi}{\hbar} |E_T|^2 V_T \cdot \frac{\sqrt{2}}{\pi^2 \hbar^3} \sqrt{m_{\perp}^*} m_{0\parallel} \sqrt{E_{\text{trap}} - E_{c0}} \frac{1}{\exp\left(-\frac{E_{\text{trap}} - E_{Fi}}{kT}\right) + 1} \quad \text{[TC}_j\text{]}$$

j^{th} segment along total V_{sol} segments

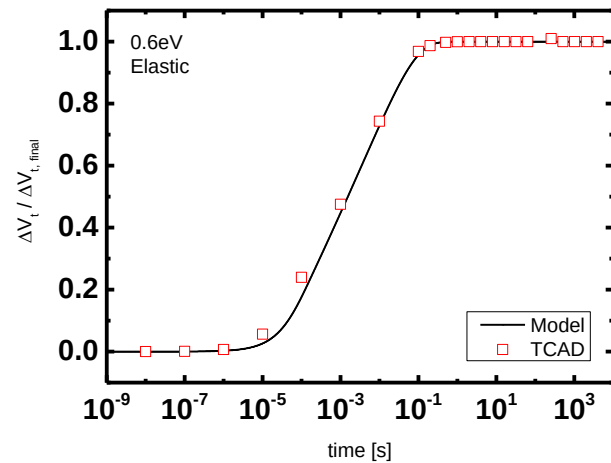
$$\begin{aligned} \text{[TC}_j\text{]} &= \exp\left(-\frac{4\sqrt{2m_n^*} E_T^{1.5} - \left(E_T - \frac{(j-\lambda)}{V_{\text{sol}}} F_n t_{n1}\right)^{1.5}}{3\hbar q F_{n,j}}\right) \Rightarrow \textcircled{1} \\ &\times \exp\left(-\frac{4\sqrt{2m_{\text{ox}}^*} \left(E_T + \Delta E_c - \frac{(j-\lambda)}{V_{\text{sol}}} F_n t_{n1}\right)^{1.5} - \left(E_T + \Delta E_c - \frac{(j-\lambda)}{V_{\text{sol}}} F_n t_{n1} - F_{\text{ox}} t_{\text{tox1}}\right)^{1.5}}{3\hbar q F_{\text{ox}}}\right) \\ &\Rightarrow \textcircled{2} \end{aligned}$$



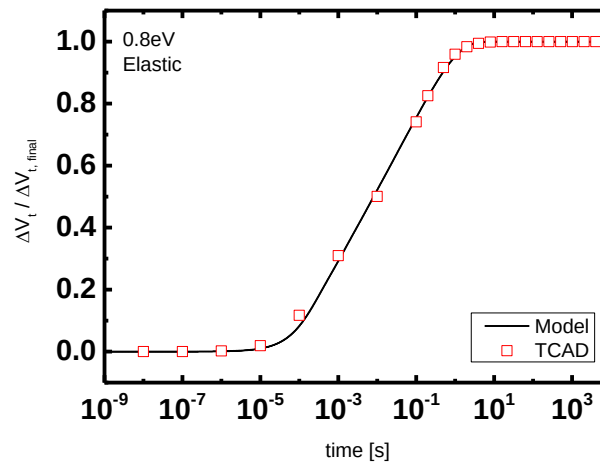
Result of Elastic Tunneling Model

Result and comparison with TCAD simulation

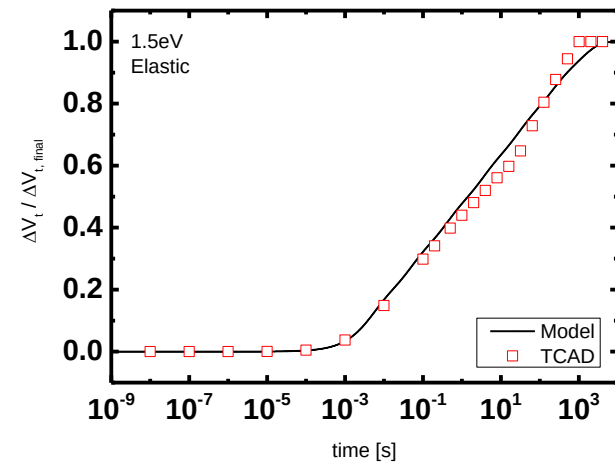
0.6 eV Elastic



0.8 eV Elastic



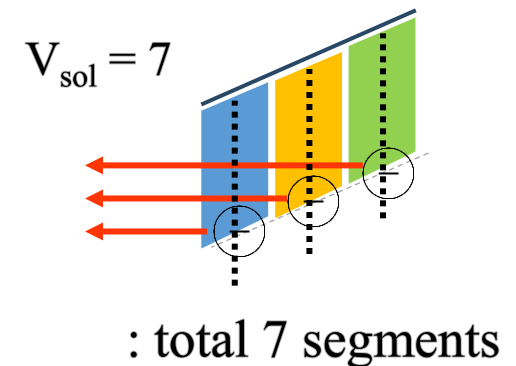
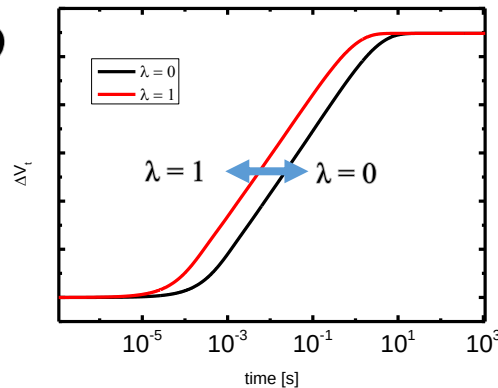
1.5 eV Elastic



Sharing parameter : $\lambda = 0.9$

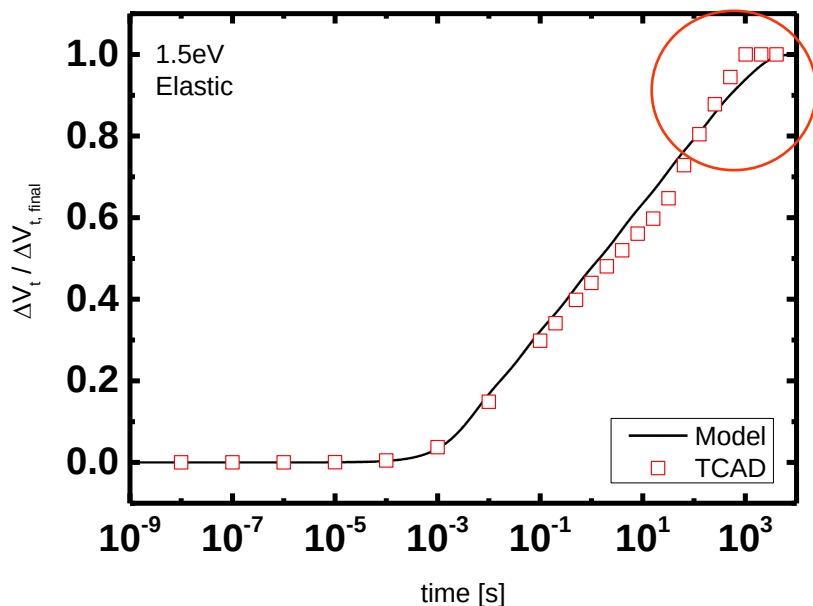


Location correction



Phonon-assisted Model

1.5 eV Elastic

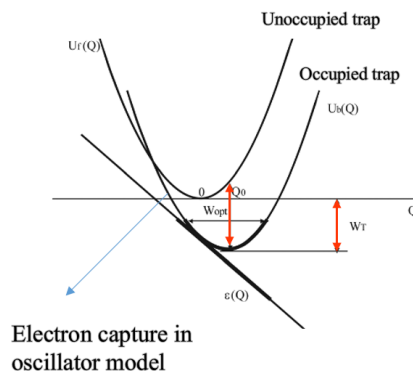


Detrap이 끝나갈 즈음, TCAD에서는
detrap 가속화

>> Phonon-assisted

$$P = \sum_{n=-\infty}^{\infty} \exp\left[\frac{nW_{ph}}{2kT} - S \coth\left(\frac{W_{ph}}{2kT}\right) I_n\left(\frac{S}{\sinh(W_{ph}/(2kT))}\right)\right] P_i(W_T + nW_{ph})$$

$$P_i(W) = \frac{eF}{2\sqrt{2mW}} \exp\left(-\frac{4\sqrt{2m}}{3\hbar eF} W^{3/2}\right)$$



W_{ph} : Phonon energy ($\hbar\omega \sim 65\text{meV}$)

W_t : Trap thermal ionization energy
= trap level

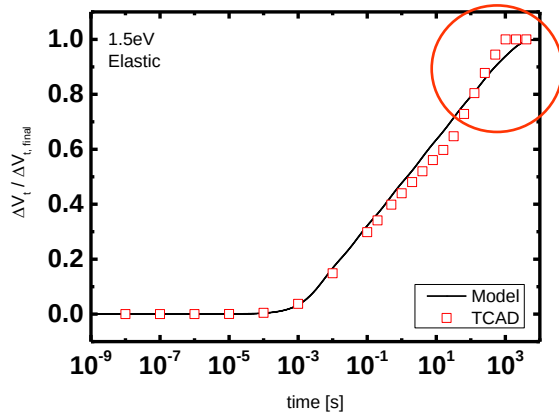
W_{opt} : Trap optical ionization energy

Q : Coordinate

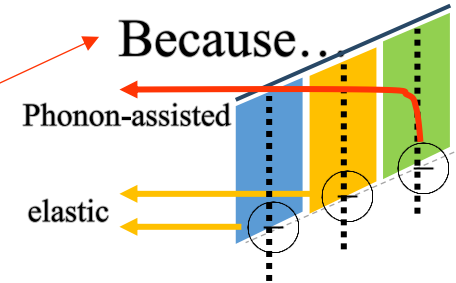
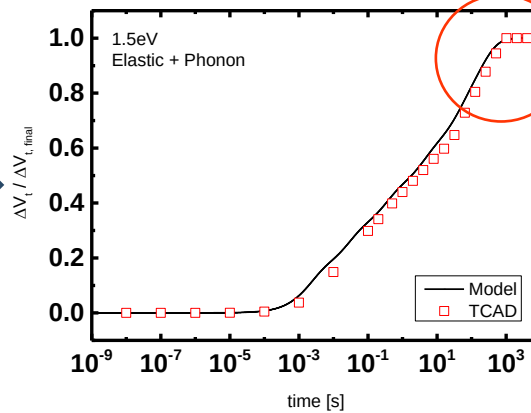
S : Huang-Rhys factor (Optical factor)

Phonon-assisted Model Result

1.5 eV Elastic



1.5 eV Phonon-assisted

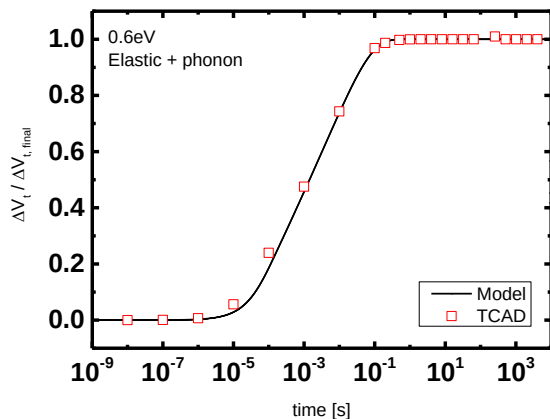


Detrapping over time

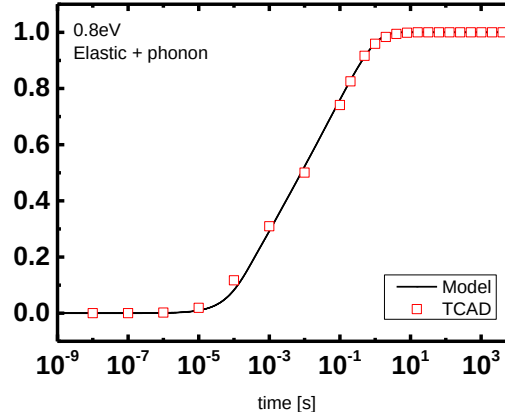
- 앞부분 부터 elastic으로 빠짐
- 남아있는 뒷부분은 Phonon-assisted

Elastic 과 거의 차이 없음

0.6 eV Phonon-assisted

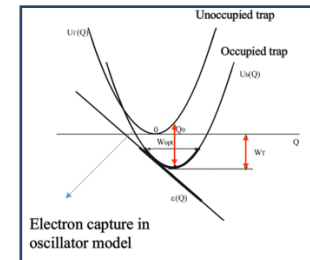


0.8 eV Phonon-assisted



$$W_{ph} = 0.06 \text{ eV}$$

$$W_t = E_t$$



W_{opt} fitting result in 2nd order

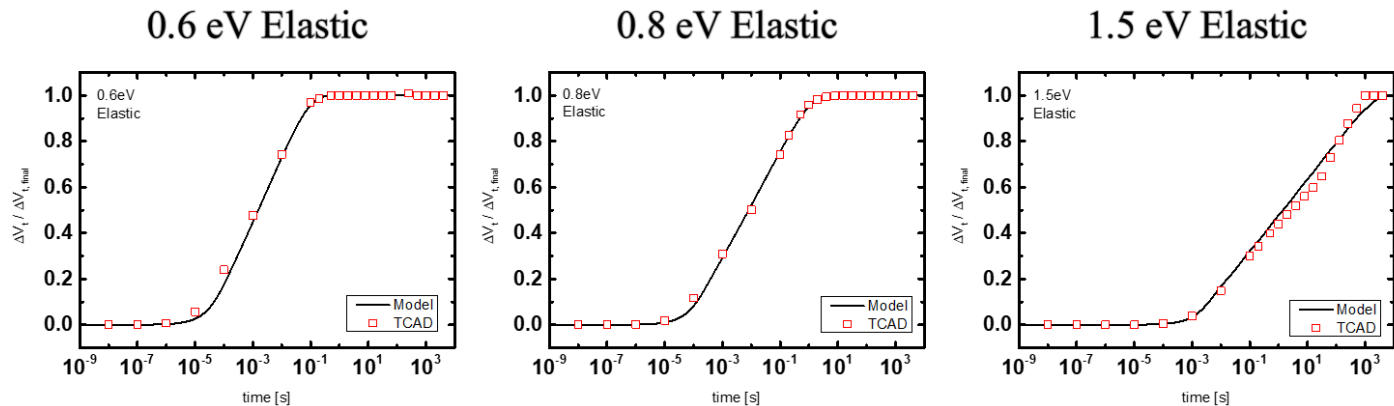
$$W_{opt} = E_t + (E_t + 0.2)^2$$

Summary

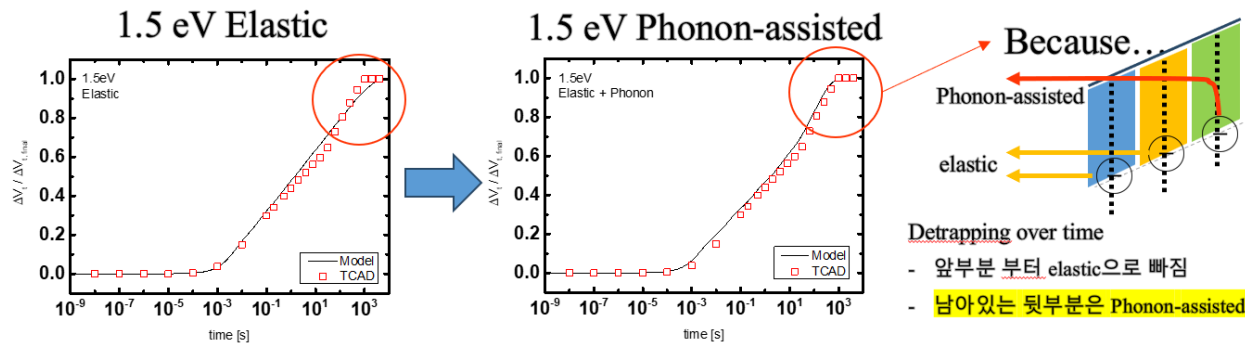
$$e_{n,elastic} = \frac{2\pi}{\hbar} |E_T|^2 V_T \cdot \frac{\sqrt{2}}{\pi^2 \hbar^3} \sqrt{m_{\perp}^*} m_{0\parallel} \sqrt{E_{trap} - E_{c0}} \frac{1}{\exp\left(-\frac{E_{trap} - E_{Fi}}{kT}\right) + 1} |TC|$$

: Trap-to-Band **elastic** tunneling emission rate

Elastic
Tunneling
TCAD Fitting



Phonon-
assisted
correction in
deep trap



Appendix

Elastic Tunneling Interface-Trap coupling

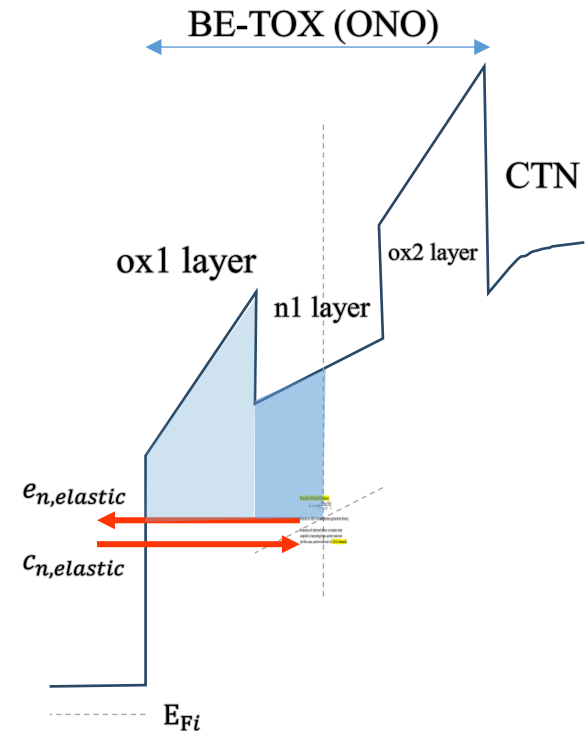
Interface-Trap Coupling by Elastic Tunneling

Principle of Detailed Balance

$$e_i = c_i \exp\left(\frac{E_{\text{trap}} - E_{Fi}}{kT}\right)$$

Similar to SRH recombination-generation theory,

Emission of electron/hole is in steady state coupled to tunneling from carrier reservoir (in this case, carrier reservoir is **CB of channel**)



$$c_{n,elastic} = D(E_{\text{trap}}) \cdot W_{\text{transition}} \cdot f_c(E_{\text{trap}}) \cdot |TC|$$

$$e_{n,elastic} = c_{n,elastic} \exp\left(\frac{E_{\text{trap}} - E_{Fi}}{kT}\right)$$

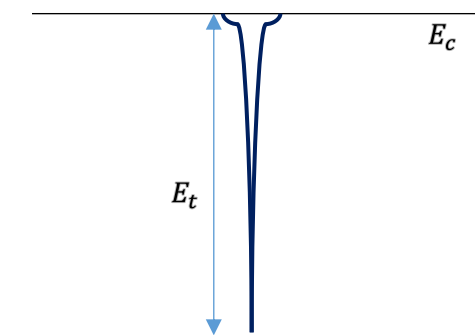
Capture Rate by Elastic Transition

$$c_{n,\text{elastic}} = D(E_{\text{trap}}) \cdot W_{\text{transition}} \cdot f_c(E_{\text{trap}}) \cdot |TC|$$

$$W_{\text{transition}} = \frac{2\pi}{\hbar} |\langle f | H' | i \rangle|^2 \delta(E - E_T) : \text{Transition rate by Fermi golden rule}$$

$$H' = \begin{cases} 0 \\ -E_T \end{cases} : \text{Constant perturbation of energy band by trap}$$

$(H_B + H')\psi = E_c \psi$: Unperturbed energy band free electron



$(H_B + H')\psi = (E_c - E_t)\psi$: Trapped electron

$$\Psi_f(\mathbf{r}) = \begin{cases} \frac{1}{\sqrt{V_T}} & \text{if } \mathbf{r} \in \text{trap} \\ 0 & \text{if } \mathbf{r} \notin \text{trap} \end{cases} \quad \text{Constant Trap}$$

$$\Psi_i(r) = \frac{1}{\sqrt{A}} \exp(i(k_x x + k_y y + k_z z))$$

Free wave eq.

$$\langle f | H' | i \rangle = E_T \frac{\sqrt{V_T}}{\sqrt{A}}$$

$$W_{\text{transition}} = \frac{2\pi}{\hbar} V_T E_T^2 \delta(E - E_T)$$

Capture/Emission of Nonlocal Tunneling

$$c_{n,\text{elastic}} = D(E_{\text{trap}}) \cdot W_{\text{transition}} \cdot f_c(E_{\text{trap}}) \cdot |\text{TC}|$$

$$= \int D_{\parallel} D_{\perp} W_C f_C = \int D_{\parallel} D_{\perp} \frac{2\pi}{\hbar} |\langle f | H' | i \rangle|^2 \delta(E - E_T) f_C dE$$

$$= \frac{2\pi}{\hbar} |E_T|^2 V_T \cdot \frac{\sqrt{2}}{\pi^2 \hbar^3} \sqrt{m_{\perp}^*} m_{0\parallel} \sqrt{E_{\text{trap}} - E_{c0}} \frac{1}{1 + \exp\left(\frac{E_{\text{trap}} - E_{Fi}}{kT}\right)} |\text{TC}|$$

$$e_{n,\text{elastic}} = c_{n,\text{elastic}} \exp\left(\frac{E_{\text{trap}} - E_{Fi}}{kT}\right)$$

$$= \frac{2\pi}{\hbar} |E_T|^2 V_T \cdot \frac{\sqrt{2}}{\pi^2 \hbar^3} \sqrt{m_{\perp}^*} m_{0\parallel} \sqrt{E_{\text{trap}} - E_{c0}} \frac{1}{\exp\left(-\frac{E_{\text{trap}} - E_{Fi}}{kT}\right) + 1} |\text{TC}|$$